

Infosys Springboard Virtual Internship 7.0

Completion Report

Team Details

- **Batch Number:** Batch 2
- **Start Date:** July 2026
- **Names:**
 - Varun Gunnala,
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 - Gummidi Sruthi,
 - Thejas S Gowda,
 - Barsha Roy,
 - Bongu Teja,
 - Bhargavi Latha Reddy,
 - Boppa Bhavyasri
- **Internship Duration:** 8 Weeks

1. Project Title

IPCMS: Integrated Patient Care Management System

2. Project Objective

The primary objective of MediTrack is to design and develop a centralized, secure, web-based Electronic Health Record (EHR) platform that overcomes the inefficiencies of traditional paper-based and fragmented digital healthcare systems. The system aims to eliminate duplicate patient records, minimize scheduling conflicts, streamline communication between patients and healthcare providers, and provide real-time access to patient health information, consultation details, and digital prescriptions. It also focuses on reducing administrative workload by organizing essential healthcare processes within a single platform. Furthermore, MediTrack ensures that sensitive patient information is protected through strict role-based access control, secure authentication, authorization, audit logging, and data security practices. By integrating these features into one unified system, MediTrack aims to improve coordination between patients, doctors, and administrators while providing a more organized, efficient, and accessible healthcare management experience.

3. Project Description in Detail

- Approach: MediTrack utilizes a modular, multi-tier software architecture following Agile development practices. It features an Analysis Engine for data normalization and risk stratification, role-based access control (RBAC), and RESTful API integration connecting the core database to provider portals, patient interfaces, and partner services.
- Technology Stack:
 - Backend & API: Java, Spring Boot, Spring Data JPA, Spring Security, JWT (JSON Web Tokens), RESTful APIs
 - Database & ORM: PostgreSQL managed via Spring Data JPA
 - API Documentation & Packaging: Swagger/OpenAPI, Docker
 - CI/CD & DevOps: GitHub Actions, Automated Testing Pipelines
 - Notification & Communication: Email, SMS, and in-app alert services
- Real-World Impact: The platform transforms healthcare administrative workflows by cutting scheduling overhead, preventing consultation delays, ensuring immediate retrieval of critical patient data during emergencies, and generating operational analytics to optimize clinical resource allocation.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, architectural design, and patient registration schema definition.	Finalized entity models, set up the development environment, and implemented basic registration forms.
Week 2	Profile management implementation, appointment slot database design, and booking workflows.	Built slot management interface, achieved appointment booking/retrieval, and validated Milestone 1 deliverables.
Week 3	Consultation module creation, clinical data capture, and doctor EHR access flows.	Implemented diagnosis recording, symptom logging, and real-time medical record updates.
Week 4	Digital prescription generator, treatment history tracking, and RBAC security rules.	Built prescription issuance engine with dosage/frequency tracking, integrated doctor-only access filters, and completed Milestone 2.

Week 5	REST API development, Swagger integration, and automated notification services.	Exposed endpoints for patient/appointment lifecycles; integrated email/SMS reminder triggers.
Week 6	Security framework integration, JWT authentication, and audit logging setup.	Implemented token-based auth, configured security event audit logs, performed load testing, and met Milestone 3 criteria.
Week 7	Analytics dashboard implementation, administrative metrics visualization, and database indexing.	Built reporting dashboards for visit trends and doctor performance; optimized database queries.
Week 8	End-to-end integration testing, Docker containerization, CI/CD pipeline setup, and documentation.	Completed 10,000-record volume testing, built GitHub Actions CI/CD workflows, drafted deployment guides, and finalized Milestone 4.

5a. Key Milestones

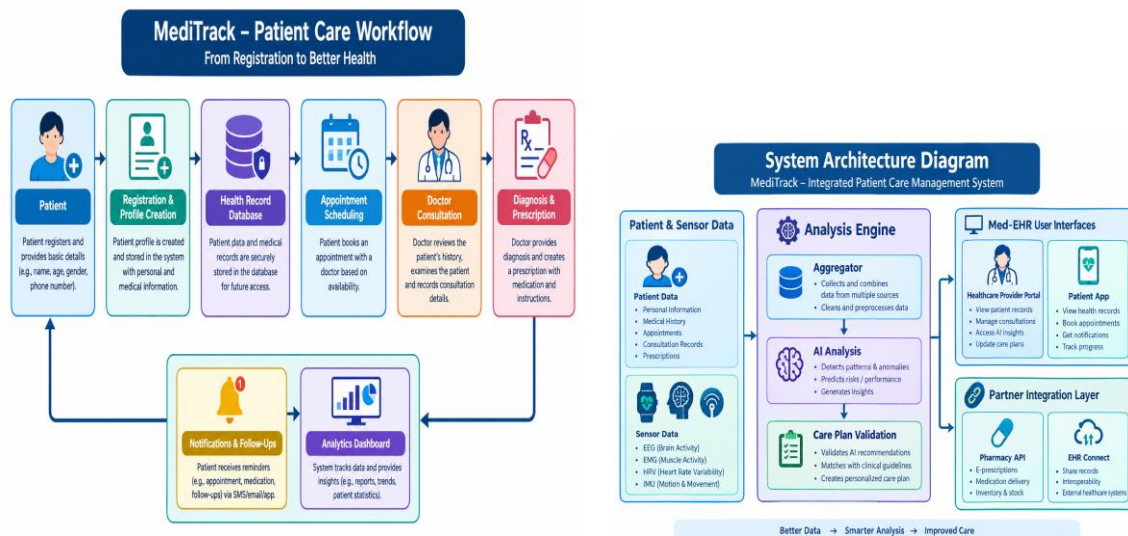
Milestone	Description	Date Achieved
Project Kickoff	Project onboarding, architecture blueprint sign-off, and environment setup.	End of Week 1
Prototype/First Draft	Patient registration, EHR profile storage, and initial appointment slot management functional.	End of Week 2
Mid-Term Review	Consultation tracker, digital prescription generator, and doctor RBAC fully integrated.	End of Week 4
Final Submission	REST APIs with Swagger, JWT security, notification alerts, analytics dashboard, Docker packaging, and CI/CD pipelines completed.	End of Week 8
Presentation	Final demonstration of end-to-end workflows, system performance review, and defense.	Project Conclusion

5b. Project execution details

The project was executed across four distinct development cycles adhering to the milestone roadmap:

1. **Milestone 1 (Weeks 1–2) – Patient Records & Basic Scheduling:** Implemented the core relational schema (Patients, Profiles, Admissions). Developed web forms and validation logic for capturing demographic, allergic, and contact data, ensuring $\geq 99\%$ data integrity alongside a conflict-free appointment scheduling interface.
2. **Milestone 2 (Weeks 3–4) – Consultation & Prescription Management:** Built provider-facing consultation interfaces allowing doctors to record symptoms, diagnoses, and lab results (Clinical_Data, Treatments). Developed a prescription generator tracking medication dosage, frequency, and duration with strict doctor-only access control.
3. **Milestone 3 (Weeks 5–6) – Notifications, Security & APIs:** Engineered RESTful endpoints documented via Swagger/OpenAPI. Configured stateless Spring Security using JWT tokens, enabled audit logging for security compliance, and built automated reminder triggers for upcoming appointments and renewals.
4. **Milestone 4 (Weeks 7–8) – Analytics, Testing & Finalization:** Constructed an administrative analytics dashboard showcasing total bookings, cancellation rates, satisfaction scores, and API latencies. Validated the system with a 10,000-record performance stress test, packaged services into Docker containers, and configured GitHub Actions CI/CD deployment pipelines.

6. Snapshots / Screenshots



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Dermatology

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Book

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ENT

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Available

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Dr. Kavya Nair

Neonatology

Paediatrics

10 yrs exp

₹500

Available

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Electrophysiology

Dr. Nisha Verma

Sports Injury

Dr. Priya Sundaram

High-risk Pregnancy

Dr. Rahul Deshpande

Joint Replacement

pg. 5

7. Challenges Faced

- **Scheduling Conflicts & Concurrency:** Race conditions during simultaneous appointment bookings were mitigated by implementing database-level transactional locks and intelligent slot allocation routines.
- **EHR Data Privacy & Role Boundaries:** Ensuring clinical files were restricted solely to attending physicians was resolved by configuring Spring Security method-level annotations (@PreAuthorize) validated against signed JWT claims.
- **High-Throughput Query Performance:** Database latency under heavy loads was addressed by creating targeted composite indexes across patient IDs, appointment timestamps, and diagnosis codes in PostgreSQL.

8. Learnings & Skills Acquired

- **Enterprise Frameworks:** Practical experience building decoupled enterprise backends using Spring Boot, Spring Data JPA, and Spring Security.
- **Authentication & Compliance:** Implemented stateless authentication patterns using JWT and structured tamper-evident audit logging for healthcare compliance.
- **Database Modeling:** Designed relational database schemas handling complex healthcare relationships (1:1 profiles, 1:N admissions, multi-type clinical records).
- **DevOps Practices:** Containerization using Docker, OpenAPI specification generation, and automated testing/deployment pipelines using GitHub Actions.

9. Testimonials from team

"Developing MediTrack provided invaluable exposure to enterprise-level architecture in the healthcare domain. Building modules that guarantee patient data integrity and handle real-time scheduling challenges bridged the gap between academic concepts and production software engineering."

10. Conclusion

The 8-week virtual internship successfully delivered the MediTrack platform, fulfilling all target functional specifications and evaluation benchmarks. By handling end-to-end healthcare administration from secure patient onboarding to automated analytics and digital prescriptions the project demonstrated scalable software engineering principles and established strong foundations for future career endeavors in full-stack enterprise development.

11. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing the platform and resources through the Virtual Internship 7.0 program. We extend special thanks to our technical mentors for their continuous guidance, architectural reviews, and constructive feedback throughout the 8-week development lifecycle.